Longitudinal Data Modeling: Run-In Power Analysis Notebook

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Contents

1 9/16/19 – Encinitas: Project Plan and Proposed Model

Plan 9/16

- Review current printed lit
- Capture key points, hopes
- Capture all useful refs in Mendeley
- Include refs in A11 paper

Josh: Move A11 → B/1. Expand text from 14-page slides. Include simulation, HMS show 28/10.

Proposed model for run-in development

As of today proposed model for run-in development is

$$Y_{ij} = \left(\beta + b_i + \beta_x g_i\right) t_j + e_{ij} \tag{1}$$

where

$$i = 1, \ldots, N \qquad j = 0, \ldots, K$$

$$b_i \sim N(0,\, \sigma_1^2) \qquad e_{ij} \sim N(0,\, \sigma^2)$$

$$g_i = \begin{cases} 1 & \text{treatment group}, \quad i = 1, \ldots, m \\ 0 & \text{placebo group}, \quad i = m+1, \ldots, N \end{cases}$$

4/16: Characterize pre-post vs longitudinal.

2 9/17/19 – Relative Efficiency: Run-In vs Standard

Compare precision of run-in design with 'R' pre-randomization observations to standard design.

To determine relative efficiency we calculate the variance with one baseline and one followup and compare to variance w/ R baselines and one followup.

Sample size formula

Model: $x_{ij} = \alpha_i + \beta_j + e_{ij}, \quad i = 1, \ldots, m, \quad j = 1, \ldots, K$

$y_{ij} = \alpha_y + \beta_j + e_{ij}$

$z_{ij} = x_{ij} - x_{i0} = y_{ij} - y_{i0}$

$$z_{\alpha/2}\sqrt{V(\bar{z}_1 - \bar{z}_2)} = z_{1-\beta}\sqrt{V(\bar{z}_1 - \bar{z}_2)} + \Delta \tag{2}$$

or

$$V(\bar{z}_1 - \bar{z}_2) = \frac{\Delta^2}{\left(z_{\alpha/2} - z_{1-\beta}\right)^2} \tag{3}$$

Assuming equal samples per group size m:

$$V(\bar{z}_1 - \bar{z}_2) = V(\bar{z}_1) + V(\bar{z}_2) = \frac{V(X_K - X_0)}{m} + \frac{V(Y_K - Y_0)}{m} \tag{4}$$

or

$$\frac{2\,V(X_K - X_0)}{m} = \frac{\Delta^2}{\left(z_{\alpha/2} - z_{1-\beta}\right)^2} \tag{5}$$

$$m = \frac{2\,V(X_{iK} - X_{i0})\left(z_{\alpha/2} - z_{1-\beta}\right)^2}{\Delta^2} \tag{6}$$

One baseline & one followup

$$\begin{aligned} V(X_K - X_0) &= 2\sigma^2 - 2\rho\sigma^2 = \sigma^2 + \sigma^2 - 2\,\mathrm{Cov}(X_K,\, X_0) \\ &= 2\sigma^2(1-\rho) \end{aligned} \tag{7}$$

R baselines, one followup

$$\begin{aligned} V(X_K - \bar{X}_0) &= V(X_K) + V(\bar{X}_0) - 2\,\mathrm{Cov}(X_K,\, \bar{X}_0) \\ &= V(X_K) + V(\bar{X}_0) - 2\rho\sqrt{V(X_K)\,V(\bar{X}_0)} \end{aligned} \tag{8}$$

$$V(X_K) = \sigma^2 \tag{9}$$

$$\begin{aligned} V(\bar{X}_0) &= V\left(\frac{X_{(K-1)} + \cdots + X_0}{K}\right) = \left(K\sigma^2 + 2\rho\,\frac{K(K-1)}{2}\,\sigma^2\right)\frac{1}{K^2} \\ &= \left(K\sigma^2 + \rho\,K(K-1)\,\sigma^2\right)\frac{1}{K^2} \\ &= \frac{\sigma^2\big(1 + \rho(K-1)\big)}{K} \end{aligned} \tag{10}$$

3 9/17/19 (cont.) – Efficiency Formula

$$
\begin{aligned}
V(X_K - \bar{X}_0) &= \sigma^2 + \frac{\sigma^2}{K}\big(1+\rho(K-1)\big) - 2\rho\sqrt{\sigma^2 \cdot \frac{\sigma^2}{K}\big(1+\rho(K-1)\big)} \\
&= \sigma^2\left[1 + \frac{1}{K}\big(1+\rho(K-1)\big) - 2\rho\sqrt{\frac{1+\rho(K-1)}{K}}\right] \\
&= \sigma^2\left[\frac{K+1}{K} + \frac{K-1}{K}\rho - 2\rho\sqrt{\frac{1+\rho(K-1)}{K}}\right] \qquad (11)
\end{aligned}
$$

So the ratio is independent of σ^2.

$$
\boxed{\text{Inv. Efficiency} = \frac{2(1-\rho)}{\dfrac{K+1}{K} + \dfrac{K-1}{K}\rho - 2\rho\sqrt{\dfrac{1+\rho(K-1)}{K}}}} \qquad (12)
$$

See report A11. Assumes constant correlation.

4 9/21 – OLS Centering Shortcut

Consider the simplest case: OLS linear regression. Looking for a centering shortcut.

Model

$$Y_i = \beta_0 + \beta_1 X_i + e_i \tag{13}$$

OLS estimates: $\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'Y$ with $X = (\mathbf{1}, \mathbf{x})$.

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \qquad \hat{\beta}_1 = \frac{\sum (X_i - \bar{X})\, Y_i}{\sum (X_i - \bar{X})^2} \tag{14}$$

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = (X'X)^{-1}$$

Centering transformation

Now transform

$$Z_i = \frac{X_i - \bar{X}}{s_x}, \qquad s_x = \sqrt{\frac{1}{n-1} \sum (X_i - \bar{X})^2}$$

Observe:

$$\sum Z_i = 0, \qquad \mathrm{Var}(Z_i) = 1, \qquad \sum Z_i^2 = n - 1$$

We can now work with Z_i to obtain closed form expressions for $\hat{\boldsymbol{\beta}}_z = (\hat{\beta}_{1z}, \hat{\beta}_{2z})$ and $V(\hat{\boldsymbol{\beta}}_z)$.

Transformed model

$$Y_i = \beta_{0z} + \beta_{1z}\, Z_i + e_i \tag{15}$$

$$V(\hat{\boldsymbol{\beta}}_z) = (Z'Z)^{-1} = \begin{pmatrix} \frac{1}{n} & 0 \\ 0 & \frac{1}{n-1} \end{pmatrix} \tag{16}$$

$$\hat{\boldsymbol{\beta}}_z = (Z'Z)^{-1}Z'Y = \begin{pmatrix} \bar{Y} \\ \dfrac{\sum Z_i Y_i}{n-1} \end{pmatrix} \tag{17}$$

Back-transformation to X scale

Thus given $\hat{\beta}_{1z}$ we can transform back to X with

$$\hat{\beta}_{1x} = \hat{\beta}_{1z} \left[\frac{(n-1)\, \hat{s}_x}{\sum X_i^2} \right] = \hat{\beta}_{1z} \left[\frac{1}{s_x} \right] \tag{18}$$

$$\mathrm{Var}(\hat{\beta}_{1x}) = \mathrm{Var}(\hat{\beta}_{1z}) \left[\frac{(n-1)\, \hat{s}_x}{\sum X_i^2} \right]^2 = \mathrm{Var}(\hat{\beta}_{1z}) \left[\frac{1}{s_x} \right]^2 \tag{19}$$

5 9/25/19 – Mixed Effects Model

$$y_{ij} = \beta_0 + b_{0i} + (\beta_1 + b_{1i} + \beta_x g_i)\, t_j + e_{ij} \tag{20}$$

Differencing from baseline:

$$\begin{aligned} z_{ij} = y_{ij} - y_{i0} &= (\beta_1 + b_{1i} + \beta_x g_i)(t_j - t_0) + e_{ij} - e_{i0} \\ &= (\beta_1 + b_{1i} + \beta_x g_i)\, d_j + \varepsilon_{ij} \end{aligned} \tag{21}$$

where $d_j = t_j - t_0$ and $\varepsilon_{ij} = e_{ij} - e_{i0}$.

In matrix form:

$$\mathbf{z}_i = X_i \boldsymbol{\beta} + V_i b_i + \boldsymbol{\varepsilon}_i \tag{22}$$

$$b_i \sim N(0,\, D), \qquad \boldsymbol{\varepsilon}_i \sim N(0,\, R)$$

Let $w_i = \frac{1}{s_d}(d_i - \bar{d})$, $\quad s_d^2 = \frac{1}{n-1}\sum (d_i - \bar{d})^2$.

GLS estimation

From GLS (Thm?):

$$\hat{\boldsymbol{\beta}}_{s_d} = (X^{*\prime} V^{*-1} X^*)^{-1}\, X^{*\prime} V^{*-1} X^* \tag{23}$$

where X^* is X with w replacing d.

$$\begin{aligned} \boldsymbol{\beta} &= (\beta_0,\, \beta_x) & b_i &= (b_{1i}) \\ X_i &= (\mathbf{d},\, \mathbf{d})\ [\text{trt}] \ = \ (\mathbf{d},\, \mathbf{0})\ [\text{plc}] & D &= \sigma_b^2 \\ & & R &= \sigma^2 I \end{aligned}$$

$$\mathrm{Var}(\hat{\boldsymbol{\beta}})\, s_d^2 = (X^{*\prime} V^{*-1} X^*)^{-1}$$

Solution steps

1. Calculate $V_i^* = R_i + U_i' D\, U_i'$
2. V_i^{*-1}
3. $X_i' V_i^{*-1} X_i$ for $g_i = 1$ and $g_i = 0$
4. $F = \sum_{\text{trt}} X_i' V_i^{*-1} X_i + \sum_{\text{plc}} X_i' V_i^{*-1} X_i$
5. F^{-1}
6. $F^{-1}_{(2,2)} = V(\hat{\beta}_x^*), \quad V(\hat{\beta}_x) = V(\hat{\beta}_x^*)\, \dfrac{1}{s_d^2}$

6 9/28/19 – Debugging: Closed-Form $\hat{\boldsymbol{\beta}}$

Long log E-date 9-28-19.

Short term problem

Trying to find closed form expressions for $\hat{\boldsymbol{\beta}}$ with model

$$\begin{aligned} y_{ij} &= (\beta_1 + b_i + \beta_x g_i)\, t_j + e_{ij} \qquad \text{[check w/ notes]} \\ &= X_i \boldsymbol{\beta} + U_i b_i + e_{ij} \end{aligned} \tag{24}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_x \end{pmatrix} \qquad \text{key is } \beta_x \text{ (interaction)}$$

GLS theory says:

$$\hat{\boldsymbol{\beta}} = (X'V^{-1}X)^{-1}\, X'V^{-1}\mathbf{y} \qquad \text{in ML/REML/univ?}$$

I worked out algebra and got

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} \sum w_i \bar{y}_i \\ 0 \end{pmatrix} \qquad \text{[check]}$$

Issue: $\hat{\beta}_x = 0$ doesn't seem right.

How to solve? Strategies

1. Rework algebra
2. Calculate $(X'V^{-1}X)^{-1}\, X'V^{-1}\mathbf{y}$ for small data set
3. <u>Construct X, V, $\mathbf{y}$ in Mathematica and calculate/simplify</u>

2a) Extract R_i, D from GLS output: $\quad R_i = \sigma^2 I, \quad D = \sigma_1^2$

7 11/11 – Generalized Model with Phase Indicators

$$Y_{ij} = \alpha + a_i + \big(\delta(1-h_j) + \beta h_j + \gamma h_j g_i + b_i\big)\, t_i + e_{ij} \tag{25}$$

$$h_j = \begin{cases} 0 & \text{if } j \le r \\ 1 & \text{if } j > r \end{cases} \qquad b_i \sim N(0,\, \sigma_b^2), \qquad e_{ij} \sim N(0,\, \sigma^2)$$

Change scores

$$c_{ij} = Y_{ij} - Y_{i0}$$

$$c_{ij} = \big(\delta(1-h_j) + \beta h_j + \gamma h_j g_i + b_i\big)\, t_i + (e_{ij} - e_{i0}) \tag{26}$$

Run-in:

$$\begin{aligned} c_{i1} &= Y_{i,-r} - Y_{i,0} \\ &\vdots \\ c_{ir} &= Y_{i,-1} - Y_{i,0} \end{aligned}$$

Fixed followup:

$$\begin{aligned} c_{i,r+1} &= Y_{i,1} - Y_{i,0} \\ &\vdots \\ c_{i,r+k} &= Y_{i,K} - Y_{i,0} \end{aligned}$$

Common close:

$$\begin{aligned} c_{i,r+k+1} &= Y_{i,K+1} - Y_{i,0} \\ &\vdots \\ c_{i,r+k+f} &= Y_{i,K+f} - Y_{i,0} \end{aligned}$$

$$g_i = \begin{cases} 1 & \text{if trt} \\ 0 & \text{if plac} \end{cases}$$

Idea: Add more slope partitions to model "Leber" designs.

8 Derivation Plan: $\mathrm{Var}(\hat{\gamma})$ Under Various Conditions

Plan

Derive $\mathrm{Var}(\hat{\gamma})$ under conditions:

0. $r = 1,\ f = 1$
1. $r = 0,\ f = 0$
2. $r = 2,\ f = 0$
3. $r = 2,\ f = 2$

Case 0

$$c_{i1} = Y_{i,-1} - Y_{i,0} = -(\delta + b_i)\, t_j + e_{i,1} \tag{27}$$
$$c_{i2} = Y_{i,1} - Y_{i,0} = (\beta + \gamma g_i + b_i)\, t_j + e_{i,2} \tag{28}$$

$$\mathrm{Var}(c_{i,1}) = \sigma^2, \qquad \mathrm{Var}(c_{i,2}) = \sigma^2, \qquad \text{let } t_j = t$$

$$\begin{aligned}\mathrm{Cov}(c_{i,1},\, c_{i,2}) &= \mathrm{Cov}(Y_{i,-1},\, Y_{i,1}) - \mathrm{Cov}(Y_{i,-1},\, Y_{i,0}) - \mathrm{Cov}(Y_{i,0},\, Y_{i,1}) \\ &\quad + \mathrm{Cov}(Y_{i,0},\, Y_{i,0}) \\ &= 0 + 0 + 0 + \sigma^2 = \sigma^2 \end{aligned} \tag{29}$$

$$G = \sigma_b^2$$
$$R = \begin{pmatrix} 2\sigma^2 & \sigma^2 \\ \sigma^2 & 2\sigma^2 \end{pmatrix}, \qquad R^{-1} = \frac{1}{3\sigma^2}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \tag{30}$$

$$X\boldsymbol{\beta} + Z\mathbf{b} = \begin{pmatrix} -t & 0 & 0 \\ 0 & t & \\ 0 & t & g_i t \end{pmatrix} \begin{pmatrix} \delta \\ \beta \end{pmatrix} + \begin{pmatrix} t \\ t \end{pmatrix} (b_i) \tag{31}$$

Woodbury identity

$$\boxed{X'V^{-1}X = X'R^{-1}X - X'R^{-1}Z\left(G^{-1} + Z'R^{-1}Z\right)^{-1} Z'R^{-1}X} \tag{32}$$

9 Matrix Algebra: Two-Period Case

$$\begin{aligned}\left(G^{-1}+Z'R^{-1}Z\right)^{-1} &= \left[\frac{1}{\sigma_b^2}+\frac{t^2}{3\sigma^2}\left(-3,\,3\right)\begin{pmatrix}-1\\1\end{pmatrix}\right]^{-1}\\ &= \left(\frac{1}{\sigma_b^2}+\frac{2t^2}{\sigma^2}\right)^{-1} = \left(\frac{\sigma^2+2t^2\sigma_b^2}{\sigma^2\sigma_b^2}\right)^{-1}\\ &= \frac{\sigma^2\sigma_b^2}{\sigma^2+2t^2\sigma_b^2}\end{aligned} \tag{33}$$

$$X'R^{-1}X = \begin{pmatrix}-t & 0\\ 0 & t\\ 0 & tg\end{pmatrix}\frac{1}{3\sigma^2}\begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}\begin{pmatrix}-t & 0 & 0\\ 0 & t & tg\end{pmatrix} = \frac{t^2}{3\sigma^2}\begin{pmatrix}2 & 1 & g\\ 1 & 2 & 2g\\ g & 2g & 2g^2\end{pmatrix} \tag{34}$$

$$X'R^{-1}Z = \frac{t^2}{3\sigma^2}\begin{pmatrix}3\\3\\3g\end{pmatrix} \tag{35}$$

$$X'V^{-1}X = X'V^{-1}X\big|_{g=1} + X'V^{-1}X\big|_{g=0} \tag{36}$$

$$X'V^{-1}X = \frac{t^2}{3\sigma^2}\left[\begin{pmatrix}4 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 2\end{pmatrix} - \frac{3\sigma_b^2t^2}{\sigma^2+2\sigma_b^2t^2}\begin{pmatrix}2 & 2 & 1\\ 2 & 2 & 1\\ 1 & 1 & 1\end{pmatrix}\right] \tag{37}$$

Substitutions

Let $a=\sigma^2, \quad b=\sigma_b^2$.

$$\begin{aligned} c &= 2a+bt^2 & d-c &= bt^2-a\\ d &= a+2bt^2 & 2c-d &= 3a\\ & & 2d-c &= 3bt^2\end{aligned}$$

$$\boxed{r=\frac{a-bt^2}{t^2b+2a} \quad \longleftarrow \text{ correlation between } c_{i1},\, c_{i2}} \tag{38}$$

$$V = 2GZ'+R = \begin{pmatrix}2a+bt^2 & a-bt^2\\ a-bt^2 & 2a+bt^2\end{pmatrix} \tag{39}$$

10 Matrix Algebra (cont.): Inverse and Variance of $\hat{\boldsymbol{\beta}}$

$$r = \frac{c-d}{c} \qquad\qquad r^2 = \frac{d^2 - 2cd + c^2}{c^2} \tag{40}$$

$$1 - r^2 = \frac{d(2c-d)}{c^2}, \qquad \frac{2}{t^2}\,(c)(1-r^2) = \frac{2}{t^2}\,\frac{d(2c-d)}{c} = \frac{2d}{c}\cdot 3a$$

$$\frac{3bt^2}{a+2bt^2} = \frac{3bt^2}{d} = \frac{2d-c}{d} = 2 - \frac{c}{d}$$

$$(X'\Sigma^{-1}X)^{-1} = \left[\frac{t^2}{3a}\left(\begin{pmatrix} 4 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 2 \end{pmatrix} - \frac{2d-c}{d}\begin{pmatrix} 2 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}\right)\right]^{-1} \tag{41}$$

$$= \frac{3a}{t^2}\left(\frac{2d}{c}\right) = 2\left(b + \frac{2a}{t^2}\right)(1-r^2) \tag{42}$$

11 8/10/24 – Confirming Frost et al. (2008), Section 3.1.1

Project: Confirm calculations in Frost et al. 2008, Sections 3.1.1 and 3.1.2.

3.1.1 Two groups

$i = 1, 2. \quad g_i = \begin{cases} 0 & \text{if placebo} \\ 1 & \text{if treat} \end{cases}$

Observations at: Y_{i0}, $Y_{i,1}$, $Y_{i,2}$ at times $(0,\, t,\, 2t)$.

Treatment effect: alter rate of change.

Model

$$Y_{ij} \sim N\big[\alpha + (\beta + \gamma g_i)\,\tau_j + a_i + b_i\, t_j;\ \sigma^2\big] \tag{43}$$

$$\begin{pmatrix} a_i \\ b_i \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix},\ \begin{pmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{pmatrix}\right]$$

$$\boxed{\begin{aligned} &E(Y|w) = X\beta + Zw \\ &V(Y|w) = R \\ &V(w) = G \\ &V(Y) = \Sigma = R + 2GZ' \end{aligned}}$$

Useful device

Recognize that $c_{ij} = Y_{ij} - Y_{i(j-1)}$ depends only on β and γ (α falls out).

$$\begin{pmatrix} c_{i1} \\ c_{i2} \end{pmatrix} \sim N\left[\begin{pmatrix} \beta + \gamma g_i \\ \beta + \gamma g_i \end{pmatrix} t + \begin{pmatrix} b_i \\ b_i \end{pmatrix} t\,,\ \begin{pmatrix} 2\sigma^2 & -\sigma^2 \\ -\sigma^2 & 2\sigma^2 \end{pmatrix}\right] \tag{44}$$

$$b_i \sim N(0,\, \sigma_b^2), \qquad R = \sigma^2 \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad R^{-1} = \frac{1}{3a}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

GLS estimator

Consider 2-person RCT: $g = 0$ and $g = 1$. Calculate fixed effects estimates and variances.

$$\hat{\boldsymbol{\beta}} = (X'\Sigma^{-1}X)^{-1}\, X'\Sigma^{-1}Y$$

$$\sqrt{\hat{\boldsymbol{\beta}}} = (X'\Sigma^{-1}X)^{-1}$$

$$\Sigma = R + 2GZ', \qquad \Sigma^{-1} = R^{-1} - R^{-1}Z\,(G^{-1} + Z'R^{-1}Z)^{-1}\, Z'R^{-1} \quad \text{(Woodbury identity)}$$

$$X_i = \begin{pmatrix} 1 & g_i \end{pmatrix} t, \qquad Z = \begin{pmatrix} t \\ t \end{pmatrix}, \qquad G = \sigma_b$$

Let $a = \sigma^2, \quad b = \sigma_b^2$.

12 8/10/24 (cont.) – Section 3.1.1 Calculations

Need for calculation: $X'R^{-1}X, \quad X'R^{-1}Z, \quad Z'R^{-1}Z, \quad X'R^{-1}Y, \quad Z'R^{-1}Y, \quad (G^{-1}+Z'R^{-1}Z)^{-1}$.

$$X'R^{-1}X = \sum_{i=1}^{2} X'_{(i)}R^{-1}X_{(i)} = \sum \frac{2t^2}{a}\begin{pmatrix} g_i \\ 1 \end{pmatrix}\begin{pmatrix} g_i & 1 \end{pmatrix} = \frac{2t^2}{a}\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \tag{45}$$

$$Z'R^{-1}Z = \frac{2t^2}{a} \tag{46}$$

$$X'R^{-1}Z = \sum \frac{2t^2}{a}\begin{pmatrix} 1 \\ g_i \end{pmatrix} = \frac{2t^2}{a}\begin{pmatrix} 2 \\ 1 \end{pmatrix} \tag{47}$$

Let c_{ij} denote data. Then:

$$X'R^{-1}Y = \frac{t}{a}\begin{pmatrix} c_{11}+c_{12}+c_{21}+c_{22} \\ c_{21}+c_{22} \end{pmatrix} \tag{48}$$

$$Z'R^{-1}Y = \frac{t}{a}\big(c_{11}+c_{12}+c_{21}+c_{22}\big) \tag{49}$$

$$(G^{-1}+Z'R^{-1}Z)^{-1} = \frac{ab}{a+2bt^2} \tag{50}$$

$$X'R^{-1}Z\,(G^{-1}+Z'R^{-1}Z)^{-1}\,Z'R^{-1}Y = \frac{2bt^3}{a(a+2bt^2)}\begin{pmatrix} c_{11}+c_{12}+c_{21}+c_{22} \\ c_{21}+c_{22} \end{pmatrix} \tag{51}$$

$$\begin{aligned} X'\Sigma^{-1}Y &= X'R^{-1}Y - X'R^{-1}Z\,(G^{-1}+Z'R^{-1}Z)^{-1}\,Z'R^{-1}Y \\ &= \frac{t}{a}\begin{pmatrix} c_{11}+c_{12}+c_{21}+c_{22} \\ c_{21}+c_{22} \end{pmatrix} - \frac{2bt^3}{a(a+2bt^2)}\begin{pmatrix} c_{11}+c_{12}+c_{21}+c_{22} \\ c_{21}+c_{22} \end{pmatrix} \\ &= \frac{t}{a+2bt^2}\begin{pmatrix} c_{11}+c_{12}+c_{21}+c_{22} \\ c_{21}+c_{22} \end{pmatrix} \end{aligned} \tag{52}$$

$$X'\Sigma^{-1}X = \frac{2t^2}{a+2bt^2}\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \tag{53}$$

$$\boxed{(X'\Sigma^{-1}X)^{-1} = \frac{a+2bt^2}{2t^2}\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \mathrm{Var}(\hat{\boldsymbol{\beta}})} \tag{54}$$

$$\boxed{(X'\Sigma^{-1}X)^{-1}\,X'\Sigma^{-1}Y = \frac{1}{2t}\begin{pmatrix} c_{11}+c_{12} \\ c_{21}+c_{22}-(c_{11}+c_{12}) \end{pmatrix} = \hat{\boldsymbol{\beta}}} \tag{55}$$